RATING METHODOLOGY

12 September 2025

TABLE OF CONTENTS

Scope 1
Sector overview 2
Rating approach 3
Scorecard 4
Discussion of the scorecard factors 6
Other considerations 9
Scorecard-indicated outcome 12
Assigning issuer-level and instrument-level ratings 13
Key rating assumptions 13
Limitations 14
Moody's related publications 15

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Rating Methodology

Steel

This rating methodology replaces the *Steel* methodology published on 12 November 2021. We have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Leverage and Coverage factor, we have replaced (CFO - Dividends)/Debt with RCF/Net Debt. We have changed some factor weights and scoring thresholds. We have also made editorial changes throughout the methodology to enhance readability, and we have changed the presentation of the scorecard.

Scope

This rating methodology applies to companies globally that are primarily engaged in the production and sale of carbon, stainless and specialty steel on a semi-finished or value-added basis, including integrated steel producers and mini-mills.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

This methodology also applies to producers of downstream fabricated aluminum products or other metal-based products whose revenue and earnings are derived from a margin-on-metal construct, in which a company converts a metal into a higher-margin product.

We rate companies that have activities in the above-mentioned areas, but that are primarily engaged in the extraction or mining of ore, coal or other minerals, using our methodology for mining companies.[1]

This rating methodology was republished on 16 October 2025 to correct the Aaa values for the EBIT Margin and EBIT/Interest Expense subfactors in the scorecard. We have also corrected the values in footnotes 1 and 6 of the scorecard and made edits to scorecard presentation for style and consistency.

Sector overview

The steel industry is mature, cyclical and exhibits a high degree of earnings volatility. Industry fundamentals are influenced by general global economic conditions and are sensitive to the performance of industries that use steel, such as the automotive, construction, energy and durable-equipment industries, among others.

Steel prices are volatile, reflecting the limited pricing power of producers of the commodity-like metal as well as supply-demand imbalances. Pervasive global overcapacity and disparities between local and international steel prices can lead to increased trading activity that exacerbates this volatility. Because of these imbalances, the steel industry has become the focus of significant political and regulatory scrutiny across jurisdictions with substantial production capacity. This scrutiny can take the form of trade cases or protective domestic policies against steel imports.

Raw material costs also can be volatile, especially the cost of scrap metal, iron ore and coking coal, and disconnects between movements in steel prices and input costs can squeeze margins. The steel industry typically has significant operating leverage, particularly among integrated producers. The ability to run plants at optimal utilization levels is critical to absorbing fixed costs and to achieving or maintaining cost-competitiveness despite the price volatility at both the input and output ends of the business.

Additionally, the industry remains vulnerable to substitution risk in key markets such as the automotive industry, where it has experienced increasing penetration by aluminum to lightweight vehicles. In most cases, a steel company's ability to introduce technologically advanced products, such as high-strength steel, is critical to its long-term viability.

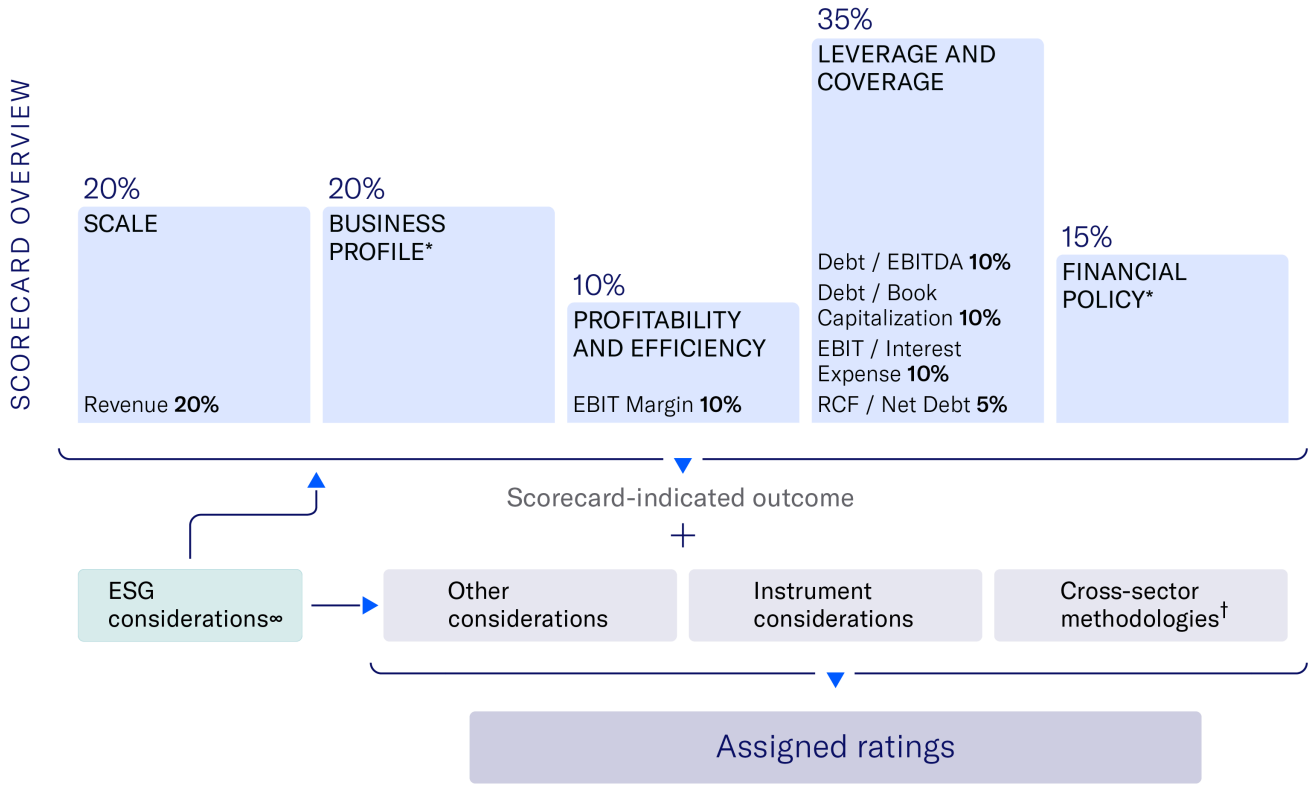
Although most of these challenges are common among steel producers, business characteristics may vary significantly. The type of steel a company produces, as well as its end markets, scale, cost-competitiveness, technological capabilities, and the strength of its customer relationships, are key differentiating factors among companies and can drive resilience to industry risks.

Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1
Illustration of the steel rating approach



* This factor has no subfactors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Steel scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
Factor: Scale (20%)										
Revenue (USD Billion) [1]	20%	≥ $100	$50 - $100	$25 - $50	$10 - $25	$4 - $10	$1.5 - $4	$1 - $1.5	$0.5 - $1	< $0.5
Factor: Business Profile (20%)										
Business Profile	20%	Essentially no earnings volatility is expected; commanding market position with well-balanced global exposure and no vulnerability to imports or cyclicality; a very high percentage of value-added products across extremely diverse end-markets; entrenched ability to control costs with technological advantages; and geopolitical environment poses no risk of impact on commercial activities.	Very low earnings volatility is expected; deeply entrenched and leading market position with global exposure and highly defensible position against imports and cyclicality due to cost-effectiveness and technological competencies; a high percentage of value-added products across highly diverse end-markets; and geopolitical environment poses very low risk of impact on commercial activities.	Low earnings volatility is expected; strong market position in its relevant markets with a high percentage of value-added products; demonstrated and sustainable competitive advantages with modest vulnerability to imports and insulation from fluctuations in raw material costs; very broad diversity in products, end-markets and geographies; and geopolitical environment poses low risk of impact on commercial activities.	Moderate earnings volatility is expected; solid market position in its most important geographic or product markets; good mix of value-added and commodity products, helping to mitigate vulnerability to imports; a competitive cost position or can pass through the majority of its costs; good diversity in products, end-markets and geographies provides a buffer against sudden, unexpected shifts in demand; and geopolitical environment poses modest but manageable risk of impact on commercial activities.	Somewhat elevated earnings volatility is expected; products are largely undifferentiated or more commodity-based, or the marketplace is highly competitive, exposing the company to periods of heightened volatility and to a greater vulnerability to imports; however; this exposure is tempered by an established market position or by a favorable cost position or by an ability to pass through raw material costs or by fair diversity characteristics but still-modest concentration in products, end-markets or geographies; or geopolitical environment poses moderate risk of impact on commercial activities.	Substantial earnings volatility is expected; products are undifferentiated, or commodity products form a significant percentage of output; or competition is intense and customers are price-sensitive, making results highly volatile; or highly vulnerable to imports or no advantageous cost profile; or limited technological capabilities or other mitigating competitive advantage; or high operational concentration; or geopolitical environment poses high risk of impact on commercial activities.	Extremely volatile earnings are expected; modest market presence or mostly produces commodity products; or has few competitive advantages; or has above-average costs or limited technological capabilities; or very high operational concentration (1 or 2 locations); or geopolitical environment poses very high risk of impact on commercial activities.	Near-term earnings are difficult to predict with any degree of confidence; no material competitive advantage; or extremely high operational concentration; or geopolitical environment poses extremely high risk of impact on commercial activities.	Near-term earnings are expected to be extremely negative with no expectation of recovery; no material competitive or cost advantage.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
Factor: Profitability and Efficiency (10%)										
EBIT Margin % [2]	10%	≥ 25%	20% - 25%	15% - 20%	10% - 15%	7% - 10%	4% - 7%	1% - 4%	0% - 1%	<0%
Factor: Leverage and Coverage (35%)										
Debt / EBITDA [3]	10%	≤ 0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4x	4x - 6x	6x - 7.5x	7.5x - 9x	>9x
Debt / Book Capitalization [4]	10%	≤ 20%	20% - 30%	30% - 40%	40% - 50%	50% - 70%	70% - 80%	80% - 90%	90% - 110%	> 110%
EBIT / Interest Expense [5]	10%	≥ 20x	15x - 20x	10x - 15x	5x - 10x	3x - 5x	1.5x - 3x	1x - 1.5x	0.5x - 1x	<0.5x
RCF / Net Debt [6]	5%	≥ 70%	50% - 70%	40% - 50%	25% - 40%	15% - 25%	8% - 15%	1% - 8%	0% - 1%	<0%
Factor: Financial Policy (15%)										
Financial Policy	15%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.	Expected to have financial policies (including risk and liquidity management) that create imminent risk of debt restructuring.

[1] For the linear scoring scale, the Aaa endpoint value is $150 billion. A value of $150 billion or better equates to a numeric score of 0.5. The C endpoint value is zero. A value of zero equates to a numeric score of 21.5.
[2] For the linear scoring scale, the Aaa endpoint value is 40%. A value of 40% or better equates to a numeric score of 0.5. The C endpoint value is (15)%. A value of (15)% or worse equates to a numeric score of 21.5.
[3] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The C endpoint value is 12x. A value of 12x or worse equates to a numeric score of 21.5, as does a negative Debt/EBITDA value.
[4] For the linear scoring scale, the Aaa endpoint value is 0%. A value of 0% equates to a numeric score of 0.5. The C endpoint value is 140%. A value of 140% or worse equates to a numeric score of 21.5, as does a negative Debt/Book Capitalization value.
[5] For the linear scoring scale, the Aaa endpoint value is 30x. A value of 30x or better equates to a numeric score of 0.5. The C endpoint value is (1)x. A value of (1)x or worse equates to a numeric score of 21.5.
[6] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The C endpoint value is (5)%. A value of (5)% or worse equates to a numeric score of 21.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[2] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and a company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (20% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

Core aspects of a steel company's business profile include its market position, cost-competitiveness, product mix and diversity of end markets as well as its susceptibility to geopolitical events, all of which drive resilience to volatility.

A steel company's market position is important because it influences its susceptibility to volatility and its long-term sustainability. Product quality, manufacturing expertise, technical know-how and customer service have a large degree of influence on a company's market position because they help build defensible customer relationships, strengthen pricing power or lock in long-term delivery contracts.

Steel producers typically have little ability to differentiate their products and mainly compete on the basis of price, over which they exert little control, therefore making low costs one of their most significant competitive advantages. With raw material costs representing a large part of total costs (often between 60% and 80%), a company's ability to control its costs is critical.

A diverse mix of production locations, products, end markets and geographies is important because it reduces a company's exposure to one product or developments in one economy that could cause earnings to erode. Having operations in multiple locations lessens the impact of strikes, equipment failures, power outages and other operational event risks that could significantly curtail output. Production concentration at one or a few mill sites creates elevated operational risk and gives a company fewer options to optimize

production. Steel and fabricated aluminum producers with broad customer bases in the automotive, construction, home appliance and manufacturing industries are often able to better cope with demand weakness in a particular customer industry. In addition, geographic diversity helps steel and downstream aluminum companies mitigate the impact of trade barriers and tariffs on their exports.

Because the trading of steel products and the raw materials used for steel production is global, geopolitical events in countries that are major producers of steel, iron ore and metallurgical coal are another important consideration because such events could have a profound impact on the business of steel companies around the globe. For example, suspended production of iron ore or coal in one region could disrupt the supply chain of steel companies in another region. A government that seeks to reduce overcapacity or step up environmental requirements could constrain local production to the benefit of foreign producers. Conversely, import restrictions, tariffs and anti-dumping provisions typically benefit local producers at the expense of producers selling into that country and may have spillover effects to other markets.

How we assess it for the scorecard

We score this factor based on our qualitative assessment of the durability of demand and a company's ability to meet that demand.

We generally do not expect a given company to match each attribute listed for a given scoring category for this factor. We typically assign a factor score based on the alpha category for which a company has the greatest number of characteristics. However, there may be cases in which one characteristic is sufficiently important to a particular company's credit profile that it has a large influence on the factor score. For example, a reliance on a small number of operating locations is likely to limit the factor score to the B or Caa rating category, because operating risk in these cases is typically very high. Strikes, equipment failure, power outages and other operating problems are more likely to affect companies whose output and profitability depend on one, or only a few, mill sites. Such companies are also more likely to have limited product and end-market diversity, thereby negatively affecting their earnings stability.

We consider a steel company's market position and ability to offer value-added products or differentiated services to its customers; its ability to manage fluctuations in raw material costs and to pass through cost inflation to customers; its diversity in terms of operations, product offerings, end markets and geographies; and the impact of geopolitical risk on its business activities.

Our assessment of a steel company's market position is typically based on its share of the market for its major products, its pricing power and its ability to secure long-term contracts with customers by offering value-added products or differentiated services. We also may consider potential changes in its competitive landscape as a result of market consolidation, new entrants and technological progress.

In assessing a company's ability to control costs, we typically consider the extent to which a company's raw material procurement policy and inventory management, as well as its bargaining positions with major suppliers, insulate it from the effects of raw material price movements. Backward integration in raw materials, whereby a steelmaker merges with or acquires a former supplier, may be another consideration because it can lead to greater cost control and often translates into higher profitability compared with peers.

The protection may be enhanced by contractual agreements that enable a steel company to pass through certain costs to customers. This feature has been more common in mature economies, such as the US and Japan, and in certain product markets, such as automotive, and stainless steel and specialty alloys. The markets for iron ore and coking coal generally function either on a contract basis or on a spot, monthly or quarterly price basis in the seaborne markets. Depending on where prices are at any point in time, steel companies that operate on a shorter-term basis can be exposed to volatility in costs, while companies that operate on a contract basis can ultimately end up with prices that are above or below the market.

We may also consider a company's ability to reduce its operating leverage, which is typically high in the steel and downstream aluminum industries. This may also support flexibility and ability to respond to temporary reductions in demand for steel or aluminum products. Additionally, producers that are able to reduce their energy consumption are also more cost-competitive and able to lower their carbon intensity.

In assessing a company's business profile, we generally consider several dimensions of diversity. We may consider concentration risk in terms of production locations and end markets as well as product types and customer segments. Companies with a balanced contribution from a number of production locations, products or end markets typically score higher for this factor because of the

protection diversity affords against the risk of disruption in a specific market. Limited operational diversity can constrain scoring for this factor, particularly if the company relies heavily on one or two operating locations.

Our assessment of geographic diversity typically considers the breadth of a company's operations and sales internationally. However, we likely would not view broad diversification by itself as an advantage if the company does not also have meaningful business scale, product leadership or cost advantages in the areas in which it operates.

In assessing a company's exposure to geopolitical risk, we typically consider risks that emanate from regulation (including changing regulation), government actions and major events in any jurisdiction that could negatively affect the commercial activities of steel or downstream aluminum companies in that jurisdiction or in other jurisdictions. Such risks include tightening environmental regulations, policies to reduce excess steel capacity, suspended mining activities, import or export restrictions and tariffs, and anti-dumping provisions.

In addition, we typically consider the impact of local or international military conflicts on a steel producer. We may assess the materiality of geopolitical risks in light of the likelihood of a particular event and the extent of its impact on a company.

Factor: Profitability and Efficiency (10% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBIT Margin.[3]

EBIT Margin

We use the ratio of earnings before interest and taxes (EBIT) to revenue.

Factor: Leverage and Coverage (35% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on four quantitative subfactors: Debt/EBITDA; Debt/Book Capitalization; EBIT/Interest Expense and RCF/Net Debt.[4]

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

Debt / Book Capitalization

We use the ratio of total debt to book capitalization.

EBIT / Interest Expense

We use the ratio of earnings, before interest and taxes (EBIT) to interest expense.

RCF / Net Debt

We use the ratio of retained cash flow (RCF) to net debt.

Factor: Financial Policy (15% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[5]

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[6]

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics that are in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity

stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[7]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Seasonality

Seasonality can be an important driver of customer demand and can cause swings in cash balances and working capital positions for some steel companies. Higher volatility creates less room for errors in product or operational execution.

Cyclical sectors

In a cyclical sector, the scorecard-indicated outcome may be higher than the assigned rating at the top of the economic cycle and lower than the assigned rating at the bottom of the cycle. While using annual or other 12-month-period financials in the scorecard typically provides useful insights into recent or near-term financial results, ratings may also reflect our expectations for the progression of financial results over a longer period that may include a full economic cycle. However, cyclicality itself poses many different types of risks to companies, and cycles do not reverse themselves with predictable regularity. A secular decline or expansion may also affect a cyclical sector. Ratings may incorporate these considerations qualitatively.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
1	3	6	9	12	15	18	20	21

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4

Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5	20.5 - 21.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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